

Impact of the MW-M31 Merger on the Dark Matter Halo Distribution of M33

NICOLAS MAZZIOTTI¹

¹ *University of Arizona*

Keywords: Local Group, Cold Dark Matter Theory, Dark Matter Halo, Halo Shape, Oblate/Prolate/Triaxial

1. INTRODUCTION

The majority of a galaxy’s mass is believed to come from its **dark matter halo**, a cloud-like structure of non-luminous matter that envelops the visible components of a galaxy. Existence of the dark matter halo was first hypothesized to explain the unusually high rotation speeds in the outskirts of nearby spiral galaxies, implying that there was significantly more matter present in these galaxies than could be observationally detected (F. Zwicky 1933; V. C. Rubin & W. K. Ford 1970). Although dark matter particles are invisible and weakly-interacting, the three dimensional distribution of the dark matter halo around a galaxy—known as the **halo shape**—is not considered to be random. Geometrical conventions such as **oblate** (elongated along poles), **prolate** (flattened at poles), or **triaxial** (asymmetric poles) matter distributions are commonly used metrics for quantifying the halo shape, based off of analyses of observational and simulated data.

Dark matter and baryons are the fundamental building blocks of a **galaxy**. Relating the dark matter halo shape to observable gas and stellar features in galaxies across cosmic time is essential for studying **galaxy evolution**, which correlates the morphology of a galaxy to a specific evolutionary stage. This paradigm forms the basis of the Lambda **cold dark matter** (Λ CDM) **theory** of cosmology, which asserts that dense regions of dark matter in the early universe provided the necessary scaffolding for galaxy formation by pulling in sufficient amounts of baryonic matter. Galaxies in the **Local Group**, namely the Milky Way (MW), M31, and M33, are ideal candidates to study dark matter halos because of their proximity, especially the halos of satellites that would otherwise be too distant to probe. This may lend valuable insight into a longstanding issue in Λ CDM cosmology: the missing satellites problem, which arises from the fact that fewer satellites have been found in

the Local Group than predicted (G. Kauffmann et al. 1993).

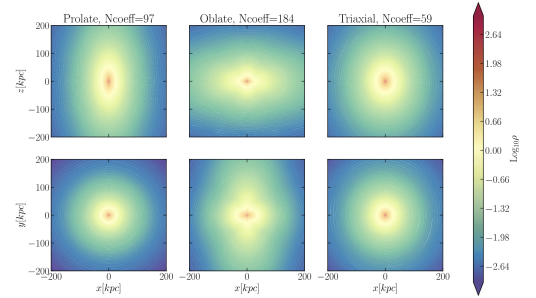


Figure 24. Density contours in the y - z and x - z planes of the prolate, oblate and triaxial halos are shown. The coefficients were computed up to $n_{\text{max}} = l_{\text{max}} = 20$ and the $\Gamma = 5$ was used to truncate the expansion. For the oblate halo the expansion needs high-order terms to reproduce the density field.

Figure 1. Figure 24 taken from N. Garavito-Camargo et al. (2021) displaying the morphology of prolate (left column), oblate (middle column), and triaxial (right column) dark matter halos. The top row is the x-z projection while the bottom row is x-y.

Efforts to characterize the 3D shape of dark matter halos are largely driven by simulations, due to the invisible nature of dark matter. By examining Milky Way sized dark matter halos in the Aquarius cosmological N-body simulations, C. A. Vera-Ciro et al. (2011) found that halo shape measured at the virial radius typically evolved from prolate to a more triaxial/oblate distribution. Narrow filaments of infalling material were found to correlate more with prolate configurations, while isotropic accretion gave rise to triaxial/oblate configurations. However, galaxy interactions may cause significant deviations from these three shapes. In N. Garavito-Camargo et al. (2021), N-body simulations of the LMC passing through the Milky Way showed that asymmetric perturbations were induced in the MW halo that did not fit a prolate, oblate, or triaxial distribution. Idealized versions of these three halo shapes computed by N. Garavito-Camargo et al. (2021) are shown in Figure 1 for reference.

The connection between dark matter subhalos and the missing satellites problem is an area of ongoing research. A thorough exploration into the geometrical evolution of satellite sized halos, as opposed to Milky Way sized halos, has yet to be conducted, though simulations of galaxies in the Local Group have grown significantly in resolution. An interesting candidate is M33, as it is the third largest galaxy in the Local Group and lacks a central bulge common for spiral galaxies. Investigating how the shape of M33's dark matter halo evolves as it orbits M31 and the eventual merger between M31 and the MW could probe the stability of such satellite galaxies. A significant shift in halo configuration could be indicative of a process that might have destabilized many satellites in the early universe.

2. THIS PROJECT

In this paper, I aim to utilize the simulation data provided by [R. P. van der Marel et al. \(2012\)](#) of the MW-M31-M33 system to characterize the shape of M33's dark matter halo from the present day until ~ 12 Gyr in the future. I will attempt to characterize the 3D halo shape of M33 by examining isodensity contours of the dark matter halo distribution in different 2D projections.

This paper seeks to address how the halo shape is impacted by the MW-M31 merger in ~ 4 Gyr, if at all, and how this agrees with the predictions made by [C. A. Vera-Ciro et al. \(2011\)](#) about the evolution of halo shape. If the distribution becomes more triaxial/oblate post-merger, then the accretion of mass onto M33 as a result of the merger may have been more isotropic. If the distribution instead becomes more prolate, then narrow filaments of mass accretion may be to blame. However, if the distribution remains the same, then a massive galaxy merger such as this may not have much of an effect on the halo shape of satellites.

Investigating how the halo shape of a M33-class satellite changes in response to a galaxy merger is important for understanding how dark matter halos evolved early in the universe when galaxy mergers were commonplace for large-scale structure formation. As a larger than average satellite, any significant changes to the halo shape of M33 can be generalized to lower mass satellites, helping to better understand satellite populations and their evolution.

3. METHODOLOGY

I rely on the data from [R. P. van der Marel et al. \(2012\)](#) produced using collisionless N-body simulations of the MW-M31-M33 system, which models the dynamics of particles belonging to the dark matter halo, disk,

and bulge components of these galaxies. The gaseous component is not included in the simulations because of its substantially low fraction compared to the stellar mass in these galaxies. The dark matter halos are initially modeled with a Hernquist (1990) profile.

Most of my methods are taken from Lab 7. I plan to use the high resolution dark matter particle data of M33 in the simulation (particle type 1) to extract information about M33's halo. First, I will use the `CenterOfMass` code to calculate the center of mass of M33 according to the dark matter particles at the desired snapshot. The dark matter particles are used to calculate the COM as opposed to the disk particles because the halo COM will likely differ from the disk particle COM. I can then calculate positions of all of the dark matter particles in the COM frame. I will then use the `RotateFrame` function to align the particle coordinates frame such that the x-y projection is a face-on view of M33, and the x-z and y-z projections are edge-on. These coordinates will be used to make a projected density plot as in Lab 7 using the `matplotlib` 2D histogram, overlaid with fitted elliptical contours produced with `photutils` (see Fig. 2). This plot should have dense contours like in Fig. 1 to see the halo shape more clearly. A box that is two times the halo scale radius of M33, $a = 25.0$ kpc (from Homework 5), in diameter will be used to standardize the projection.

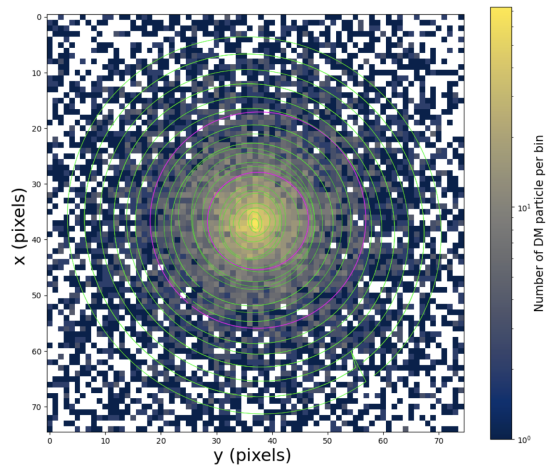


Figure 2. Face-on density projection (x-y) of the dark matter particles in M33 at Snapshot 0. Contours are overlaid in green. The outer magenta line corresponds to roughly the halo scale radius of M33 (25 kpc), while the inner magenta ring is half this scale radius.

I will attempt to classify the halo shape as either prolate, oblate, or triaxial using these fitted ellipses. I will use the ellipticity and position angle at the halo scale radius a and half this distance ($a/2$) as simple metrics

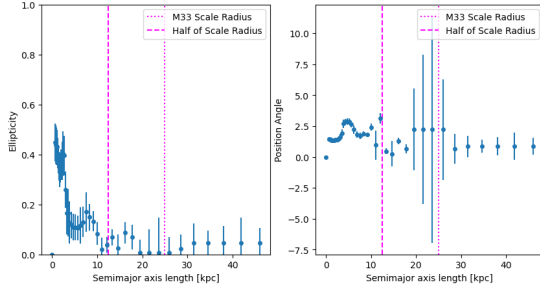


Figure 3. Corresponding fit parameters of the ellipses from Fig. 2 as a function of semimajor axis length in kpc. The left panel plot shows the ellipticity while the right plot displays the position angle. The dotted magenta line indicates the halo scale radius of M33 (25 kpc) and the dashed line is half of this.

to quantify the halo shape, examined in the x-y, y-z, and x-z projections. A prolate shape will be more elliptical at both distances with a similar position angle, whereas an oblate shape might have a position angle at $a/2$ that is perpendicular to that at a . To evaluate for triaxiality, I will use the density information to see which ellipse is closest to matching half the maximum projected density of the halo. The snapshots at which I will evaluate this ellipse information at is the present day, the end of the simulation, right before the first pericenter approach of M33 relative to M31, and then when the next apocenter

is reached after first pericenter. I will use the orbital separation plots from Homework 6 to determine these precise snapshot numbers, which are around 4 Gyr in the future, the time of the MW-M31 merger. I will create a miniature movie of the density projections during the first pericenter-apocenter range to see how the halo shape visually changes.

I hypothesize that the halo distribution of M33 will be more triaxial at the present day because it has not been recently perturbed by M31 or any other object. As the M31-MW merger occurs, I predict that the halo shape of M33 will become more prolate due to the added mass of the Milky Way, elongating M33's halo along the collision axis. I don't think the halo will deviate from a prolate, oblate, or triaxial distribution post-merger because M33 is currently far enough away from M31 and massive enough to not be completely tidally disrupted by the merger. Additionally, dark matter does not interact with itself so the halo shape should remain approximately stable despite the introduction of the MW dark matter halo during the merger.

4. RESULTS

In progress

5. DISCUSSION

In progress

REFERENCES

- Garavito-Camargo, N., Besla, G., Laporte, C. F. P., et al. 2021, *ApJ*, 919, 109, doi: [10.3847/1538-4357/ac0b44](https://doi.org/10.3847/1538-4357/ac0b44)
- Kauffmann, G., White, S. D. M., & Guiderdoni, B. 1993, *MNRAS*, 264, 201, doi: [10.1093/mnras/264.1.201](https://doi.org/10.1093/mnras/264.1.201)
- Rubin, V. C., & Ford, Jr., W. K. 1970, *ApJ*, 159, 379, doi: [10.1086/150317](https://doi.org/10.1086/150317)
- van der Marel, R. P., Besla, G., Cox, T. J., Sohn, S. T., & Anderson, J. 2012, *The Astrophysical Journal*, 753, 9, doi: [10.1088/0004-637x/753/1/9](https://doi.org/10.1088/0004-637x/753/1/9)
- Vera-Ciro, C. A., Sales, L. V., Helmi, A., et al. 2011, *MNRAS*, 416, 1377, doi: [10.1111/j.1365-2966.2011.19134.x](https://doi.org/10.1111/j.1365-2966.2011.19134.x)
- Zwicky, F. 1933, *Helvetica Physica Acta*, 6, 110